JOHN WELLARD

Senior Agentic Developer | Multi-Agent Orchestration, GraphRAG & Evaluation

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// PROFESSIONAL SUMMARY

Agentic developer shipping production multi-agent systems with deterministic orchestration, a zero-trust validator between every step, and output quality that is measured rather than assumed. My working thesis: **systems are graphs** — an LLM over flat text guesses; an LLM over a knowledge graph reasons about structure, so I model the domain in Neo4j and query it instead of hoping vector similarity lands. Shipped work includes a 12-agent orchestration engine running 755 tasks at **98.7% autonomous verification**, a hybrid GraphRAG retriever benchmarked **+18 points** over naive vector search, and edge-deployed RAG on Cloudflare Workers AI with MCP tool bridges. Behind the AI work sits **20+ years** of engineering and **15+ years** leading delivery as an **AgilePM Practitioner** — I can scope an ambiguous problem, estimate it honestly, name the delivery risks, and document it so the team can follow.

98.7%	+18pt	100%	20+
AUTONOMOUSLY VERIFIED	HYBRID RETRIEVAL LIFT	GOVERNANCE TEST COVERAGE	YEARS ENGINEERING

CORE STACK

Python · TypeScript · Claude Code · GCP Agent SDK · MCP · Neo4j / Cypher / GDS · GraphRAG · Cloudflare Workers AI · Vectorize · PostgreSQL · radCAD · Solidity · React / Next.js · GitHub Actions

// SHIPPED & RUNNING

AGENTIC WORKFLOWS

Deterministic multi-agent orchestration — poll → execute → validate → unblock, with a zero-trust validator between every step. **Agent-OS: 12 agents, 755 tasks, 98.7% autonomously verified, $0 infra.** MB governance engine: 1,345 tests at 100% coverage.

GRAPHRAG ON NEO4J

Hybrid vector + full-text retrieval over a knowledge graph — grounded and cited. **EcoGraph: 77k chunks and a 9,828-entity graph** of the full Neo4j corpus; hybrid retrieval benchmarked +18 pts with recovery doubled vs naive vector.

GRAPH-NATIVE TOKENOMICS

Token economies modelled as graphs and fused with Monte-Carlo simulation — a graph⇄sim bridge answering structural questions about dynamic output. Neo4j + GDS + radCAD, applied to live token models (DevGuild $GUILD).

RAG AT THE EDGE

Grounded retrieval on Cloudflare Workers AI + Vectorize, with MCP tool bridges so agents act on real systems safely and observably. In production across the studio edge stack (Kointel, KTHULHU).

// AI & AGENTIC ENGINEERING EXPERIENCE

Founder & AI Engineering Lead | AgileGypsy

Mar 2024 - Present • Remote

- Architected **Agent-OS**, a deterministic multi-agent orchestrator (poll → execute → validate → unblock) with a zero-trust validator between every step — 12 agents, 755 tasks, 98.7% autonomously verified, running at $0 infrastructure cost
- Designed agents that plan, decompose tasks, call tools, self-check, detect failure, retry where safe, escalate to a human where not, and **fail safely** rather than fabricating a result
- Built **GraphRAG retrieval on Neo4j** — hybrid vector + full-text over a knowledge graph (77k chunks, 9,828 entities), benchmarked +18 pts with recovery doubled against naive vector search, and answers returned grounded and cited
- Enforced strict **input/output contracts** on every tool: typed schemas, structured error shapes, schema-constrained responses — so downstream services can consume agent output without defensive parsing
- Own the **prompt engineering lifecycle** — version-controlled templates, clear system/user/tool role separation, few-shot exemplars, reproducible regression runs gating every change
- Build **evaluation pipelines** with golden datasets, regression suites, LLM-as-judge scoring and human review loops; the MB governance engine ships 1,345 tests at 100% coverage
- Implement, test and document **MCP tool bridges** so capabilities are discoverable and reusable across agents and IDE clients — Neo4j MCP tooling certified
- Deliver AI × blockchain convergence: token-gated AI SaaS platforms and autonomous on-chain agents, with tokenomics modelled as graphs in Neo4j GDS and simulated in radCAD / xyflow
- Translate client business goals into technical plans, estimates and named delivery risks; lead ambiguous research independently and bring teams along through written docs and working demos

AI Agent & Full-Stack Engineer | AgileGypsy

Jan 2026 - Present • Remote

- Ship edge-deployed RAG on **Cloudflare Workers AI + Vectorize** with MCP bridges, so agents act on real systems safely and observably (Kointel, KTHULHU)
- Build multi-step tool chains and state-machine agents with deterministic routing wrapped around the non-deterministic parts
- Instrument every LLM call with structured traces — inputs, outputs, latency, token count, tool calls, task success rate — surfaced for cost and quality regression review
- Apply prompt caching, semantic routing and output memoization to cut latency and spend while holding response consistency
- Connect agentic back ends to React / Next.js front ends over typed REST APIs with streaming responses and accessible, WCAG-conscious interaction patterns
- Stack: Python, TypeScript, Node.js, Cloudflare Workers, PostgreSQL, Neo4j, GCP Agent SDK, Docker, GitHub Actions

AI-Assisted Security Researcher | AgileGypsy / Independent

Feb 2025 - Present • Remote

- Designing a graph-native audit engine on a Code Property Graph (AST + CFG + DFG) with a multi-agent roster — Topology → Vulnerability → Exploit → Verification → Reporting — and an EVM-sandbox auto-heal loop (KTHULHU)
- Built LLM-assisted review pipelines that triage findings, draft reproduction steps and rank severity, with human verification as a mandatory gate before anything is reported
- Competitive auditing on CodeHawks, with 14 validated findings across 7 First Flights — an adversarial habit that transfers straight to red-teaming agents and prompt-injection defence

// IN DEVELOPMENT

- **Neo4j agent memory** — graph-backed long-term memory: entities, relationships and episodic recall in Neo4j rather than a flat vector store
- **AI-agent GraphRAG** — the benchmarked hybrid retriever packaged as a production agent capability over live knowledge graphs
- **Agent-ID & Web3 agent security** — identity, authorisation and security for autonomous agents acting on-chain, a fast-growing surface as agents get wallets

// TECHNICAL SKILLS

Agentic Systems & Orchestration:

Planning & task decomposition, deterministic multi-agent orchestration, zero-trust step validation, state-machine agents, multi-step tool chains, LangGraph, LangChain, AutoGen, GCP Agent SDK, custom frameworks, MCP servers & clients, function/tool calling, retry & escalation policy, guardrails, graceful failure handling

Graph & Retrieval:

Neo4j, Cypher, Graph Data Science (GDS), knowledge-graph construction & entity extraction, GraphRAG, hybrid vector + full-text retrieval, chunking & retrieval strategy, embeddings, Cloudflare Vectorize, grounded & cited answering, graph-backed agent memory

Prompt Engineering & Evaluation:

Structured prompt patterns, few-shot design, schema enforcement & output constraints, version-controlled templates, golden test sets, regression suites, LLM-as-judge scoring, human review workflows, quantitative quality metrics, prompt caching, semantic routing, output memoization

Observability & Reliability:

LangSmith, OpenTelemetry, structured logging, trace instrumentation (latency, tokens, tool calls, task success), cost tracking, canary & shadow evaluation, incident triage

Languages, Platforms & Modelling:

Python, TypeScript, JavaScript, Solidity, Rust, Node.js, React.js, Next.js, xyflow, radCAD Monte-Carlo simulation, RESTful API design, GraphQL, Cloudflare Workers, GCP, PostgreSQL, Docker, WCAG 2.2 accessibility awareness

Engineering Practice & AI Tooling:

Claude Code, GitHub Copilot, Cursor, Git & GitHub workflows, pull request review standards, CI/CD with GitHub Actions, testing practice, API contracts, coding conventions — including onboarding teammates onto AI-assisted workflows

// DELIVERY & LEADERSHIP EXPERIENCE

Project & Business Development Consultant | Independent

Oct 2022 - Mar 2024 • Remote

- Delivered 20+ projects in agile environments at 95% on-time completion and up to 40% cost savings through Lean practice
- Owned complex technical scopes end-to-end: requirements, estimates, implementation approach, risk register, stakeholder reporting
- Clients included SproutAI, Designed Bio Solutions, Thoroughbred Auto and Total Water

Project & Business Development Manager | Designed Bio Solutions

Aug 2020 - Dec 2022 • Boksburg, South Africa

- Led international engineering projects; cut delays 25% and rework costs 10% via tighter QA/QC and review gates
- 98% budget adherence with a 15% improvement in client retention

Project & Business Development Manager | Total Water Solutions

Jan 2011 - Aug 2020 • Kempton Park, South Africa

- Drove 40% revenue expansion over a nine-year tenure; expanded the client base 25% at 95% client approval
- Improved team efficiency 20% and reduced project risk exposure 20% through standardised process

Network Engineer & Lead Web Developer | Computer Concepts

Jun 2004 - Dec 2010 • Kempton Park, South Africa

- Built 200+ responsive websites and maintained production network infrastructure with an 11% uptime improvement
- Managed hosting deployments, DNS configuration and domain operations

// CERTIFICATIONS & EDUCATION

AI & Graph (Neo4j, all completed 100%):

Neo4j & Generative AI Certification | Neo4j & Generative AI Fundamentals | Developing with Neo4j MCP Tools | Neo4j Fundamentals | Neo4j Certified Professional

Delivery & Agile:

AgilePM® Practitioner | PRINCE2® Foundation | Business Agility L1 | ClickUp Expert

Engineering:

- Blockchain Development Bootcamp 3.0 • Dapp University • Jan 2026 - Jul 2026 (In Progress)
- Solidity Smart Contract Developer (SSCD+) • Cyfrin Updraft • 2025 - 2026 (In Progress)
- Chainlink Fundamentals (Jan 2026) • Cyfrin Blockchain Basics, Foundry Fundamentals, Web3 Wallet Security (2025)
- CompTIA Network+, A+ IT Technician, Microsoft MCP (2008)

Formal Education:

BSc (Honours) Environmental Science • The Open University • Part-time, in progress

> **THE THROUGH-LINE**
>
> **Systems are graphs. Model them, then prove it.** An LLM over flat text guesses; an LLM over a graph reasons about structure — who connects to whom, what depends on what, which path an attacker or a whale can take. The same discipline runs through everything I build: model the system as a graph, query it, then prove the constraints hold.